

Revised 2026-09-10 | Deliverable No. 10 | English + 简体中文

Backend Operating Manual - ellis-visa.com

Topology

Hetzner CX23 (Falkenstein). Caddy terminates TLS for ellis-visa.com and serves the static app from /opt/ellis/src/renderer/dist; /api/* is proxied to the FastAPI backend (systemd unit ellis-backend) on 127.0.0.1:8000. Data: SQLite at /var/lib/ellis/ellis.db.

Routine operations

- Status: `systemctl status ellis-backend caddy`
- Logs: `journalctl -u ellis-backend -f (backend), -u caddy (edge)`
- Deploy a tested release pinned to its commit and artifact checksum. Verify the current database backup and retain the previous release before updating. Record the release identifier, migration steps and rollback procedure; smoke test the backend, public application and quality console after restarting.
- Health: `GET https://ellis-visa.com/api/health`

Data protection

Daily full database backup is required. `backend/scripts/backup_database.py` uses SQLite's online backup mechanism to include committed WAL content, validates the destination database and atomically publishes a private (mode 0600) file only after a successful integrity check. The daily wrapper retains the existing `/var/backups/ellis/ellis-<weekday>.db` paths. Never fall back to copying a live database file when SQLite backup fails. Record successful backups and failures; verify the deployed daily wrapper and its actual output, rather than inferring success from a schedule's existence. Keep a verified off-host copy for host-loss recovery. This is a database backup; release artifacts, configuration and necessary external storage require their own recovery arrangements.

For restore, stop the service, preserve the current state, restore the verified backup using the release-specific runbook, and check database integrity and public/quality behavior before reopening service. The contractual recovery target is at most one hour; a dated timed full recovery exercise is required to prove it. `systemd` process restart is useful but is not a disaster-recovery test. The 2026-09-10 live database-only drill restored a 23,834,624-byte backup, 99 tables and 949 route-table rows with integrity ok; database backup and restore took 0.708 seconds. The daily wrapper was updated. Full-service recovery, including host/app/configuration/secrets/DNS and off-host durability, is not certified by that database check. Details are in `AVAILABILITY.md`.

Quality-control backend

<https://ellis-visa.com/#ops> - records with combined spot-check filters (names, codes and Chinese accepted), per-record 25-field checklists, confidence and source-substantiation chips, one-click error flagging into the tracked correction queue, the change log, freshness, and the two-sheet Excel export.

The current confidence display is binary High/Low, with AI/human authorship kept in source provenance and publication controlled separately by the existing evidence/conflict gates. See

`DATA_CALIBER_MANUAL.md` for precise definitions. `info_validity` is the published policy end date only; unknown dates remain blank. `freshness_valid_until` is separate internal review-deadline metadata.

The user-approved completeness target includes documented Not applicable/Not published across all 25 fields; unknown/unsupported/pending states stay incomplete. The recorded-state metric is 81.9521% on the 2026-09-10 02:29 UTC snapshot; operational applicable 20 completeness is separately 87.32%. Retain literal 25 non-null diagnostics (72.9718% cells) as well. These are completion measurements, not certification of policy accuracy.

The current bilingual HTML/PDF manuals have been regenerated for these semantics. Old PASS labels, sample availability percentages, or unmeasured recovery claims are not current acceptance evidence.

Security posture

UFW allows 22/80/443 only. Backend binds loopback; only Caddy is exposed. Secrets ship encrypted (backend/secrets.enc, AES-256-CBC) and are decrypted on the machine at deploy time; backend/.env never enters git.

Route and product publication

QC keeps every product available for inspection. Product held and route_held are distinct: a partial route can publish its supported exemption while a separate optional visa remains withheld. The public page labels this state and excludes unsupported alternative claims from its answer and chat. Conditions cannot be discarded to make a route publishable. Report default availability, fully released routes and held products separately; partial publication is not 100% product coverage.

Source: `BACKEND_OPERATING_MANUAL.md` | SHA-256
170d901d71745f4ff81140137f53c76b077899080bcef547869a96e704242ef3

Ellis 信息质量管控后台操作手册

本手册记录当前实现及操作要求，不代表全部数据已经核实或项目验收已经通过。

系统结构

生产架构使用 Hetzner CX23 (Falkenstein)，由 Caddy 为 ellis-visa.com 终止 TLS，并提供 /opt/ellis/src/renderer/dist 中的静态应用。/api/* 请求转发到本机 127.0.0.1:8000 的 FastAPI 服务 (systemd 服务名 ellis-backend)。SQLite 数据库路径为 /var/lib/ellis/ellis.db。

日常运行与发布

- 状态检查：systemctl status ellis-backend caddy。
- 后端日志：journalctl -u ellis-backend -f；边缘代理日志：journalctl -u caddy。
- 发布应使用经过测试、按提交与产物校验和固定的版本。更新前核实当前数据库备份，保留上一版本，并记录版本号、迁移步骤和回滚流程。重启后检查后台接口、公开应用与质控台。
- 健康检查地址：https://ellis-visa.com/api/health。

数据保护与恢复

每日必须进行完整数据库备份。backend/scripts/backup_database.py 使用 SQLite 在线备份接口，包含已提交的 WAL 内容；对备份执行完整性校验后，才以原子方式发布权限为 0600 的私有文件。每日执行脚本沿用 /var/backups/ellis/ellis-<weekday>.db 路径。

SQLite 备份失败时，不得退回到复制运行中的数据库文件。应记录备份成功与失败，核查线上每日任务及实际产物；任务存在不代表备份成功。应保存已验证的异机副本，以支持主机丢失后的恢复。此机制只备份数据库；发布产物、配置与必要的外部存储须分别安排恢复。

恢复时先停止服务并保留当前状态，按该版本的恢复流程还原已验证的备份，再检查数据库完整性、公开页面及质控功能，然后恢复服务。合同恢复目标为不超过 1 小时，是否达标须由带日期和计时记录的完整恢复演练证明。systemd 自动重启进程有帮助，但不是灾难恢复验收证明。2026-09-10 实际数据库备份与还原用时 0.708 秒，23,834,624 字节、99 张表、949 条线路表记录，完整性正常；每日任务已更新。该演练不包含主机、应用、配置、密钥、DNS 或异机副本可靠性，不能证明完整服务恢复达标。

质控台与数据解读

浏览器打开 <https://ellis-visa.com/#ops>。记录页面提供可组合的抽查筛选（支持名称、代码与中文）、逐条 25 字段清单、置信度与信源支持标识、问题标记、可追踪纠错队列、变更记录、时效信息及双工作表 Excel 导出。

质控保留全部产品供检查。产品 held 与线路 route_held 分别计数；部分发布的线路可以提供有证据的默认免签答案，同时拦截独立且未核实的可选签证。对客页面明确提示，并排除未核实的替代产品主张；不得为了发布而丢失入境条件。部分发布不等于 100% 产品覆盖。

当前置信度显示为 High（高）与 Low（低）两级。人工或 AI 审核身份保留在来源证据中；是否可以对客展示，由原有证据和冲突控制单独决定。详细定义见本次更新的数据口径手册。

info_validity 仅表示政策已公布的截止日期；未知日期留空。freshness_valid_until 单独表示内部复核期限。用户已同意：全部 25 字段中，有记录的不适用与未公布状态计作完整；未知、无证据或待审核仍不完整。2026-09-10 02:29 UTC 快照的记录状态完整度为 81.9521%，日常可填写 20 字段为 87.32%；全部 25 字段非空诊断为 72.9718%。这些指标不证明政策准确率。

本次版本替换旧版双语 HTML/PDF 手册。旧版 PASS 标记、示例可用率、或未计时的恢复描述，不得用作当前验收证据。

安全边界

UFW 仅开放 22/80/443 端口；后端仅绑定本机回环地址，由 Caddy 对外提供服务。部署资料中的密钥以 backend/secrets.enc 加密保存（AES-256-CBC），并在部署主机解密；backend/.env 不进入 Git。手册不包含凭据。

本文依据当前实现编制；最终验收仍需实际证据。